Labour Market Integration of Immigrants in Canada Post-Pandemic (2020–2024)

A Predictive and Industry-Level Analysis

(Literature Review, Data Description, Project Approach)

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Abstract

The COVID-19 pandemic triggered unprecedented labour market disruptions across Canada, generating widespread job displacement, sectoral volatility, and structural shifts in workforce demand. As Canada entered the post-pandemic recovery period (2020–2024), immigration levels increased in response to labour shortages and demographic pressures. Although aggregate employment indicators suggest substantial recovery, less is known about whether immigrant cohorts, particularly recent arrivals, experienced equitable labour market integration during this period.

This study examines national employment outcomes among immigrants using Statistics Canada's Labour Force Survey from 2020–2024. The analysis combines descriptive statistics with supervised machine learning classification models to evaluate employment recovery patterns, industry-level concentration disparities, and structural predictors of employment status. Logistic regression, random forest, and gradient boosting models will be implemented to assess predictive performance and feature importance.

By integrating traditional labour economics frameworks with predictive analytics, this research provides a data-driven assessment of immigrant labour market integration during the post-pandemic period. The findings aim to clarify the structural factors associated with employment outcomes without making causal displacement claims, offering evidence-based insights relevant to policy and workforce planning.

1 Literature Review

1.1 Immigrant Labour Market Integration in Canada

Research on immigrant labour market integration in Canada consistently shows that newcomers face persistent employment gaps relative to Canadian-born workers. Prior to the pandemic, recent immigrants, defined as those in Canada for fewer than five years, experienced lower employment rates, higher unemployment, and greater occupational mismatch compared to established immigrants and the Canadian-born population (Statistics Canada, 2021). These disparities are often linked to structural barriers such as non-recognition of foreign credentials, limited access to professional networks, and discrimination in hiring processes.

A substantial body of literature highlights the importance of time since arrival as a determinant of labour market outcomes. Recent immigrants typically experience the largest employment challenges, while established immigrants gradually converge toward Canadian-born outcomes as they accumulate local experience and credentials (Hou & Picot, 2016). However, even established immigrants may remain overrepresented in lower-wage or precarious occupations, which suggests that integration is influenced by systemic constraints rather than time alone.

Historical evidence shows that immigrants are disproportionately affected during economic downturns. Studies comparing the early 1990s recession, the 2008 to 2009 financial crisis, and the COVID-19 recession found that immigrants consistently experienced sharper employment declines and slower recoveries (Statistics Canada, 2022a). This vulnerability is partly due to their concentration in sectors that are more sensitive to economic shocks, such as hospitality, transportation, and certain service industries.

Early research on the COVID-19 labour market shock indicates that recent immigrants were among the hardest hit. They faced higher job losses, reduced hours, and greater instability, particularly in frontline and essential sectors (Statistics Canada, 2022b). Although employment levels began to recover by 2022, evidence suggests that the pace and distribution of recovery varied significantly across immigrant cohorts.

Overall, the literature establishes a clear pattern. Immigrant labour market integration in Canada is shaped by both structural economic forces and persistent systemic barriers. These findings provide the foundation for examining how the post-pandemic recovery unfolded between 2020 and 2024 and whether recent immigrants experienced equitable improvements in employment outcomes. However, much of the existing research focuses on pre-pandemic trends, leaving limited evidence on how these structural barriers evolved during the post-2020 recovery period.

1.2 Post-Pandemic Labour Market Recover, 2020-2024

The post-pandemic labour market recovery in Canada was uneven across industries, regions, and demographic groups. While overall employment levels returned to or exceeded pre-pandemic levels by late 2022, the underlying recovery patterns varied significantly across sectors. Statistics Canada reports that industries such as professional services, finance, and public administration recovered quickly, often surpassing pre-pandemic employment levels due to increased demand for digital services, remote-work infrastructure, and public-sector expansion (Statistics Canada, 2022b). In contrast, sectors such as accommodation, food services, transportation, and certain retail segments experienced prolonged instability.

Immigrants were disproportionately represented in many of the sectors that experienced the slowest recovery. Research shows that recent immigrants are more likely to work in frontline, customer facing, or physically demanding roles, which were more vulnerable to layoffs and reduced hours during the pandemic (Statistics Canada, 2022c). As a result, the recovery trajectory for recent immigrants lagged behind that of Canadian-born workers, even as overall employment levels improved.

Regional differences also shaped the recovery. Provinces with strong public-sector employment or technology-driven industries, such as Ontario and British Columbia, saw faster rebounds. Regions dependent on tourism or resource extraction experienced slower and more volatile recoveries. Since immigrants are not evenly distributed across provinces, these regional dynamics further contributed to uneven labour market outcomes.

Another important feature of the post-pandemic recovery was the shift in job quality. Although employment levels rose, many of the new jobs were part-time, temporary, or lower-wage positions. This shift disproportionately affected recent immigrants, who already faced barriers to accessing stable, high-quality employment. Broader national assessments also show that the pandemic's economic impacts persisted well into 2022, with immigrants continuing to experience disproportionate disruptions in employment stability and job quality during the recovery period (Statistics Canada, 2022d). As a result, employment recovery did not necessarily translate into improved economic security or upward mobility for all groups.

Overall, the literature suggests that Canada's labour market has largely recovered in aggregate terms, but the benefits of this recovery have not been evenly distributed. Immigrants, particularly recent arrivals, continue to face structural challenges that shape their employment outcomes. These findings underscore the importance of examining industry-level concentration and predictive factors influencing employment during the 2020 to 2024 period. While these studies document broad recovery patterns, they provide limited insight into how structural factors jointly predict employment outcomes for recent immigrants.

1.3 Industry-Level Segmentation & Occupations

A significant body of research shows that Canadian immigrants are unevenly distributed across industries. This pattern has important implications for employment outcomes. Industry-level segmentation refers to the tendency for certain demographic groups to be concentrated in specific sectors, often those with lower wages, higher instability, or limited opportunities for advancement. Studies consistently find that recent immigrants are more likely to work in accommodation, food services, transportation, warehousing, administrative support, and certain care-related occupations (Statistics Canada, 2021). These sectors have higher turnover rates and greater exposure to economic shocks, especially during labour market disruption.

Occupational mismatch is another well-documented challenge. Many immigrants arrive with high levels of education and professional experience, yet they often work in roles that do not align with their qualifications. Research attributes this mismatch to credential recognition barriers, limited access to regulated professions, and employer preferences for Canadian experience (Lamb, Banerjee, & Verma, 2021). Consequently, immigrants may be concentrated in sectors unable to utilize their skills, limiting long-term career mobility and earnings growth.

Industry concentration also shapes exposure to risk during economic downturns. During the COVID-19 pandemic, sectors with high immigrant representation experienced some of the largest employment declines. Accommodation and food services, transportation, and certain retail segments saw steep job losses in 2020, and recovery in these sectors was slower compared to industries such as finance, technology, and public administration (Statistics Canada, 2022b). Since recent immigrants are more likely to work in these vulnerable sectors, they faced greater instability and slower recovery trajectories.

Several studies highlight that industry segmentation is not only a short-term challenge but also a structural feature of the Canadian labour market. Even after several years in Canada, many immigrants remain concentrated in the same sectors they entered upon arrival, which suggests that occupational mobility is limited by systemic barriers rather than individual preferences alone (Hou & Coulombe, 2010). This persistent segmentation contributes to long-term disparities in earnings, job quality, and employment security.

Understanding industry-level concentration is essential for this project because it is expected to be one of the strongest predictors of employment outcomes during the post-pandemic recovery period. The literature shows that sectoral differences determined who lost work during the pandemic and who regained it during the recovery. These findings support the inclusion of industry as a key variable in the predictive modeling component of this study and provide important context for interpreting differences between recent immigrants, established immigrants, and Canadian-born workers.

1.4 Predictive Modelling in Labour Economics

Predictive modelling has become increasingly common in labour economics, particularly as researchers seek to understand complex relationships between demographic characteristics, industry factors, and employment outcomes. Traditional econometric approaches, such as linear regression and logistic regression, have long been used to analyze labour market behaviour. However, recent studies highlight the value of machine learning methods for capturing nonlinear patterns, interactions between variables, and higher dimensional data structures that are difficult to model using conventional techniques (Mullainathan & Spiess, 2017).

Machine learning models have been applied to a range of labour market problems, including predicting unemployment risk, estimating job match quality, and identifying factors associated with occupational mobility. For example, gradient boosting and random forest models have been used to predict job loss during economic downturns, and these models often outperform traditional regression approaches in terms of accuracy and sensitivity to complex patterns (Kuhn & Johnson, 2013). These methods are particularly useful when the goal is to identify the relative importance of multiple structural factors, such as education, industry, age, and region.

Despite the growing use of machine learning in labour market research, there are relatively few studies that apply these methods to immigrant labour market integration in Canada. Most existing research relies on descriptive statistics or regression-based approaches to examine employment gaps, industry concentration, or credential recognition barriers. While these studies provide valuable insights, they do not fully explore the predictive relationships between structural variables and employment outcomes during the post-pandemic recovery period.

The limited use of machine learning in this area represents an important gap. The post-pandemic labour market is characterized by rapid shifts in industry demand, uneven recovery patterns, and changing demographic dynamics. Machine learning models are well suited to capturing these complexities and can provide a more nuanced understanding of the factors that shape employment outcomes for recent immigrants. By applying supervised classification models to the Labour Force Survey, this project contributes to the emerging literature on predictive analytics in labour economics and offers a data-driven perspective on immigrant labour market integration during a period of significant economic change. However, existing research rarely integrates demographic and industry-level variables within predictive frameworks, highlighting the need for approaches that more fully capture the structural factors shaping immigrant employment outcomes during the recovery period.

1.5 Research Questions

This project is guided by the following research questions which are aligned with the project's objective of understanding immigrant labour market integration using both descriptive and predictive approaches:

1. What structural factors predict employment status for recent immigrants in Canada during the post-pandemic recovery period (2020–2024)?
2. How does industry-level concentration differ between recent immigrants, established immigrants, and Canadian-born workers?
3. How accurately can supervised machine learning models classify employment status using demographic and industry variables?

2 Data Description

The dataset selected for this project is the Labour Force Survey (LFS), a monthly household survey conducted by Statistics Canada. The LFS provides nationally representative information on labour market activity for individuals aged 15 and older who are not living in institutions or on reserves. The dataset includes detailed variables related to employment status, demographic characteristics, education, industry, occupation, and immigration background. For this project, the analysis will focus on the years 2020 to 2024 in order to capture the period of labour market disruption and recovery following the COVID-19 pandemic.

The LFS uses a rotating panel design in which households remain in the sample for six consecutive months. This structure allows for both cross-sectional and limited longitudinal analysis. Each monthly file contains approximately 50,000 households, which translates to roughly 100,000 individual respondents. When combined across five years, the dataset provides a large sample that supports robust statistical analysis and supervised machine learning.

Several variables from the LFS are directly relevant to the research questions. Employment status will serve as the dependent variable for the predictive modelling component. Independent variables will include immigrant status, time since arrival, education level, age, gender, province, and industry classification. Immigrant status is categorized by Statistics Canada into recent immigrants, established immigrants, and Canadian-born individuals, which aligns with the structure of this project. Industry is coded using the North American Industry Classification System (NAICS), which provides standardized sector definitions across years.

The dataset contains both categorical and numerical variables, which will require appropriate preprocessing. Categorical variables such as industry, education, gender, and immigrant status will be encoded using one-hot encoding or similar techniques. Numerical variables such as age will be standardized if needed. Missing values will be examined and handled using consistent rules, such as removing records with incomplete employment status or imputing values for variables with low levels of missingness. Survey weights are included in the LFS, but for the purposes of predictive modelling, the analysis will focus on unweighted microdata. Descriptive statistics will be produced to summarize the distribution of key variables across immigrant groups and across the 2020 to 2024 period.

2.1 Data Cleaning and Handling of Abnormal Values

Data cleaning will involve removing records with missing employment status, imputing low-frequency missing demographic values where appropriate, and standardizing numerical variables such as age. Categorical variables, including industry, education, gender, and immigrant status, will be encoded using one-hot encoding. Outliers and abnormal values—such as invalid age entries, inconsistent industry codes, or incomplete immigration classifications—will be identified through summary statistics and cross-tabulations. Records containing structural inconsistencies will be removed to ensure model reliability. A link to the cleaned dataset and preprocessing scripts will be provided in the project repository.

The LFS has several strengths that make it suitable for this project. It provides timely and detailed labour market information, uses consistent variable definitions across years, and includes a large sample that supports machine learning. It also contains direct measures of employment status and industry, which are central to the research questions. However, the dataset also has limitations. It does not capture informal employment, underemployment, or credential recognition outcomes. It also relies on self-reported information collected through telephone interviews, which introduces additional considerations for immigrant respondents. Recent immigrants may face language barriers that affect their ability to participate or accurately interpret survey questions. Survey participation is also more common among individuals with stable schedules, reliable phone access, and fewer competing demands. Immigrants who are newly arrived, working multiple jobs, or experiencing settlement-related stressors may be underrepresented. As a result, the dataset may slightly overrepresent more established or stable immigrant respondents, which should be considered when interpreting employment patterns.

Overall, the structure and content of the LFS support the objectives of this project. The dataset provides the necessary variables to examine employment trends among recent immigrants, assess industry-level concentration, and develop predictive models that identify the structural factors associated with employment outcomes during the post-pandemic recovery period.

3 Project Approach

The goal of this project is to examine employment outcomes for recent immigrants in Canada during the post-pandemic recovery period and to identify the structural factors that predict employment status between 2020 and 2024. The approach combines descriptive analysis with supervised machine learning in order to capture both the overall labour market patterns and the predictive relationships between demographic and industry variables.

The first step in the project will be to confirm the research questions and ensure they align with the variables available in the Labour Force Survey. The analysis will focus on employment status as the outcome of interest and will compare recent immigrants, established immigrants, and Canadian-born individuals. Descriptive statistics will be used to summarize the distribution of key variables, including industry, education, age, gender, and time since arrival. These summaries will provide an initial understanding of how different groups were positioned in the labour market during the recovery period.

The second step will involve preparing the dataset for modelling. This will include cleaning the data, handling missing values, encoding categorical variables, and selecting the features that are most relevant to the research questions. Industry, education, age, gender, province, and immigrant status will be included as predictors. The dataset will be split into training and testing sets to support model evaluation. Although the LFS includes survey weights, the predictive modelling will be conducted on unweighted microdata in order to focus on classification accuracy rather than population estimates.

The third step will be the development of supervised machine learning models. Logistic regression will serve as a baseline model because it is widely used in labour market research and provides interpretable coefficients (Mullainathan & Spiess, 2017). More flexible models, such as random forests or gradient boosting, will also be implemented to capture nonlinear relationships and interactions between variables, reflecting evidence that tree-based ensemble methods often outperform traditional regression approaches in complex labour market settings (Kuhn & Johnson, 2013). Model performance will be evaluated using accuracy, precision, recall, and the F1 score. Feature importance measures will be used to identify the variables that contribute most strongly to predicting employment status.

The final step will be to interpret the results in relation to the existing literature. The findings will be compared to prior research on immigrant labour market integration, industry segmentation, and post-pandemic recovery patterns. This comparison will help determine whether the predictive relationships observed in the data align with established trends or reveal new insights. The results will also be used to assess whether industry is a significant

predictor of employment outcomes for recent immigrants, which is a central question in this project.

Project Methodology

A visual workflow diagram to illustrate the overall methodology:

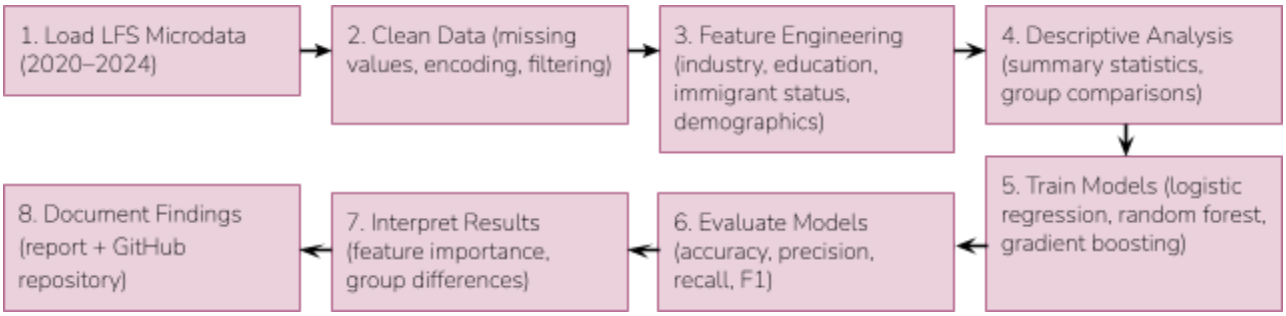

Figure 1. Project workflow illustrating data preparation, modelling, evaluation, and interpretation steps.

Project Repository

All code, data processing scripts, and model outputs will be stored in this GitHub repository:

https://github.com/jayemm2210/TMU_CIND820

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